

Theory of Computation

Exercise 5:

(Regular Grammar)

1. Give the right-linear grammar for the language
 $L = \{1^n 0^m : n \geq 1 \text{ and } m \geq 2\}.$

$$S \rightarrow 1A$$

$$A \rightarrow 1A \mid 00B$$

$$B \rightarrow 0B \mid \lambda$$

$$S \rightarrow 1S \mid 0B$$

$$B \rightarrow 0B \mid 0$$

2. Give the left-linear grammar for the language L in problem 1.

$$\begin{aligned} S &\rightarrow B 00 \\ B &\rightarrow B 0 \mid A \\ A &\rightarrow A \mid \mid \end{aligned}$$

$$\begin{aligned} S &\rightarrow S 0 \mid A 0 0 \\ A &\rightarrow A \mid \mid \end{aligned}$$

(Homework 4)

*3. Draw NFA for the following grammar G.

$G: A \rightarrow xyB,$

$B \rightarrow yxC,$

$C \rightarrow xB \mid yy$

